Recovering exponential accuracy in Fourier spectral methods involving piecewise smooth functions with unbounded derivative singularities [1]

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Abstract

Fourier spectral methods achieve exponential accuracy both on the approximation level and for solving partial differential equations (PDEs), if the solution is analytic. For linear PDEs with analytic but discontinuous solutions, Fourier spectral method produces poor pointwise accuracy, but still maintains exponential accuracy after post-processing [7]. In this paper, we develop a technique to recover exponential accuracy from the first N Fourier coefficients of functions which are analytic in the open interval but have unbounded derivative singularities at end points. With this post-processing method, we are able to obtain exponential accuracy of spectral methods applied to linear transport equations involving such functions.

Keywords: Spectral method; exponential accuracy; Fourier coefficients; Gegenbauer expansion; transport equation; singular initial conditions.

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1 Introduction

In this paper, we are concerned with the accuracy of spectral methods when applied to linear transport equations involving piecewise smooth functions with unbounded derivative singularities:

$$\begin{cases} u_t + cu_x = 0, & x \in [-1,1], \qquad t > 0 \\ u(x,0) = a(x) + b(x)(1+x)^s, & x \in [-1,1] \end{cases} \tag{1.1}$$

with periodic boundary conditions, where c is the phase speed (for simplicity, assume that $c > 0$), s is a given fractional constant

$$0 < s = \frac{p}{q} < 1 \tag{1.2}$$

with relatively prime integers p and q, and $a(x)$ and $b(x)$ are both analytic functions. The initial condition is $C^{0,\alpha}$ Hölder continuous with $\alpha \leq s$ and 2-periodic. It has a singularity in $[-1,1]$, which means its first derivative blows up at $x = -1$. The solution to this problem is

$$u(x,t) = u(x - ct, 0) \tag{1.3}$$

The singularities will move along the lines through points $(x,t) = (2n-1, 0)$ (here, $n \in Z$) with direction $(c, 1)$. Even though we consider functions with one singularities in this paper, our technique can be easily applied to functions with finitely many such singularities in $[-1,1]$ of known locations.

We start by recalling the Fourier approximation operator P_N to an L^2 function $u(x)$:

$$P_N u(x) = \sum_{|k| \leq N} \tilde{u}_k e^{ik\pi x} \tag{1.4}$$

where the Fourier coefficients $\tilde{u}_k$ are defined by

$$\tilde{u}_k = \frac{1}{2} \int_{-1}^{1} u(x) e^{-ik\pi x} dx \tag{1.5}$$

for Fourier Galerkin methods.

The approximation error

$$u(x) - P_N u(x) = \sum_{|k| > N} \tilde{u}_k e^{ik\pi x} \tag{1.6}$$

is exponentially small, if $u(x)$ is periodic and analytic. If $u(x)$ is an analytic but non-periodic function, it has a discontinuity at the boundaries of the interval, when treated as a periodic function. The truncated Fourier series has large oscillations near the jump, and the overshoot does not decay as the number of terms retained in the series increase. The approximation error is only first order in smooth regions away from the discontinuity. This is known as the Gibbs phenomenon. In [8, 3, 6, 4, 5], Gottlieb et al. developed a general framework to overcome this difficulty, in the sense that exponential accuracy is recovered in the maximum norm for any sub-interval of analyticity, from the knowledge of either the first N spectral expansion coefficients, or the point values at N standard collocation points. This means that exponential accuracy is recovered at all points, including at the actual discontinuity points (the left and right limits at these points), if the locations of these discontinuity points are known. If the locations of these discontinuity points are not known exactly but are known to be within certain fixed intervals, then exponential accuracy can be recovered from any interval which does not overlap with these fixed intervals containing the discontinuities. An important tool used in this framework is the set of Gegenbauer polynomials, which are orthogonal in the interval $[-1, 1]$ with the weight $(1 - x^2)^{\lambda-\frac{1}{2}}$. It turns out that the main ingredient in this technique is that the parameter λ in the weight function as well as the number of terms m retained in the Gegenbauer expansion should both be chosen proportional to N. We refer to [7] for a review of this series of work.

In [2], we extended this methodology to handle spectral collocation methods for functions which are analytic in the open interval but have singularities at end-points. The reconstruction procedure is performed on functions of the following form

$$f(x) = a(x) + b(x)(1 + x)^s, x \in [-1, 1] \tag{1.7}$$

where s is a given fractional constant as in (1.2), and $a(x)$ and $b(x)$ are both analytic functions. With this extension, exponential accuracy can be obtained from standard collocation point values of such functions, by properly choosing the parameters λ and m to be linearly dependent on N.

These methods are widely used as "reconstruction" or "post-processing" techniques to recover exponential accuracy for point values based on the Fourier approximation $P_N u(x)$.

In this study, we are interested in Fourier Galerkin methods for solving linear, hyperbolic time-dependent partial differential equations (PDEs) [9]. Consider the problem

$$\begin{cases} u_t - \mathscr{L}u = 0, & x \in [-1,1], \qquad t > 0 \\ u(x,0) = g(x), & x \in [-1,1] \end{cases} \tag{1.8}$$

where $\mathscr{L}$ is a differential operator. In the Fourier Galerkin method, we seek solution $u_N(x,t)$ from the space $\hat{B}_N \in \text{span}\{e^{ik\pi x}\}_{|k|\leq N}$, i.e.:

$$u_N(x,t) = \sum_{|k|\leq N} a_k(t) e^{ik\pi x} \tag{1.9}$$

where, $a_k(t)$ are unknown coefficients which will be determined by the method. In general, the coefficients $a_k(t)$ are not equal to the Fourier coefficients $\tilde{u}_k$; they will be equal only if we obtain the exact solution of the problem. In the Fourier Galerkin method, $a_k(t)$ are determined by requiring that the residual

$$R_N(x,t) = \frac{\partial u_N(x,t)}{\partial t} - \mathscr{L}u_N(x,t) \tag{1.10}$$

is orthogonal to $\hat{B}_N$. The method is defined by the requirement that the orthogonal projection of the residual onto the space $\hat{B}_N$ is zero. If the residual is smooth enough, this requirement implies that the residual itself is small.

When spectral method is used to solve PDEs, we will estimate the errors in the maximum norm. Due to the Gibbs phenomenon,

$$\|u(x) - P_N u(x)\|_{L^\infty}$$

can not be small even for $t = 0$. We would like to reconstruct the solution, and recover exponentially decaying error

$$\|u(x) - Q_N P_N u(x)\|_{L^\infty}$$

where Q_N is a post-processing operator.

The objective of postprocessing is to extract the hidden information from the truncated Fourier Series (1.9) and recover exponential accurate point values at every point including at the singularities. The reconstruction procedure is performed on functions of the form (1.7). As in [2], we assume that the analytic functions $a(x)$ and $b(x)$, denoted generically as $v(x)$, satisfy the following condition.

Assumption 1.1 *There exists a constant $\rho \geq 1$ and a constant $C(\rho)$ such that, for every $k \geq 0$,*

$$\max_{-1 \leq x \leq 1} \left| \frac{d^k c(x)}{dx^k} \right| \leq C(\rho) \frac{k!}{\rho^k}.$$

This is a standard assumption for analytic functions, where ρ is the distance from the interval $[-1,1]$ to the nearest singularity of the function $c(x)$ in the complex plane.

We will use the following one to one transformation between $x \in [-1,1]$ and $y \in [-1,1]$:

$$(2^{q-1}(1+x))^{\frac{1}{q}} = 1 + y \tag{1.11}$$

where q is defined in (1.2).

The function $\bar{f}(y) = f(x(y))$ of the variable y has its usual Gegenbauer expansion under the basis $\{C_l^\lambda(y)\}$:

$$f(x(y)) = \bar{f}(y) = \sum_{l=0}^{\infty} \hat{f}^\lambda(l) C_l^\lambda(y)$$

with the Gegenbauer coefficients $\hat{f}^\lambda(l)$ given by

$$\hat{f}^\lambda(l) = \frac{1}{h_l^\lambda} \int_{-1}^{1} (1-y^2)^{\lambda - \frac{1}{2}} \bar{f}(y) C_l^\lambda(y) dy. \tag{1.12}$$

where the precise value of the normalization constant h_l^λ will be given by (2.4) in Definition 2.1.

Our goal is to find a good approximation to the first $m \sim N$ Gegenbauer coefficients $\hat{f}^\lambda(l)$ in (1.12), denoted as $\hat{g}^\lambda(l)$ (defined later in (3.1)), from the given Fourier coefficients.

We will then obtain the approximation of $f(x)$ using these $m \sim N$ terms of its Gegenbauer expansion:

$$f_N^{m,\lambda}(x) = \sum_{l=0}^{m} \hat{g}^\lambda(l) C_l^\lambda(y(x))$$

We will separate the analysis of the error from reconstruction into two parts: the truncation error and the regularization error. The truncation error measures the difference between the exact Gegenbauer coefficients of $f(x(y))$ with $\lambda \sim N$, and the approximate Gegenbauer coefficients $\hat{g}^{\lambda}(l)$ obtained by using the truncated Fourier Series. This will be investigated in Section 3. The regularization error measures the difference between the Gegenbauer expansion with $\lambda \sim N$, using the first $m \sim N$ Gegenbauer coefficients, and the function itself. This error is estimated in Section 4. The results for the reconstruction are summarized in Theorem 4.3 in Section 4. The analysis of the error in the Fourier Galerkin method with reconstruction applied to the initial boundary problem (1.1) will be illustrated in Section 5. Section 6 contains several numerical examples to illustrate our results. In Section 2, we shall give several useful preliminary properties and estimates. Concluding remarks are given in Section 7.

Throughout this paper, we will use C to denote a generic constant either independent of the growing parameters, or depending on them at most in polynomial growth. The details will be indicated clearly in the text. These constants may not take the same value at different places.

2 Preliminaries

In this section, we will introduce the Gegenbauer polynomials and discuss some of their asymptotic behavior. Then we will give some estimates as preparation for the error estimates in Section 3.

Definition 2.1 *The Gegenbauer polynomial $C_n^{\lambda}(x)$, for $\lambda \geq 0$, is defined by*

$$(1-x^2)^{\lambda-\frac{1}{2}} C_n^{\lambda}(x) = \frac{(-1)^n}{2^n n!} G(\lambda, n) \frac{d^n}{dx^n}\left[(1-x^2)^{n+\lambda-\frac{1}{2}}\right]$$

where $G(\lambda, n)$ is given by

$$G(\lambda, n) = \frac{\Gamma(\lambda+\frac{1}{2})\Gamma(n+2\lambda)}{\Gamma(2\lambda)\Gamma(n+\lambda+\frac{1}{2})}. \tag{2.1}$$

for $\lambda > 0$*, by*

$$G(0,n) = \frac{2\sqrt{\pi}(n-1)!}{\Gamma(n+\frac{1}{2})}$$

for $\lambda = 0$ *and* $n \geq 1$*, and by*

$$G(0,0) = 1$$

for $\lambda = 0$ *and* $n = 0$*. Notice that by this standardization,* $C_n^0(x)$ *is defined by (see [1]):*

$$C_n^0(x) = \lim_{\lambda \to 0^+} \frac{1}{\lambda} C_n^\lambda(x) = \frac{2}{n} T_n(x), \qquad n > 0; \qquad C_0^0(x) = 1,$$

where $T_n(x)$ *are the Chebyshev polynomials.*

Under this definition we have, for $\lambda > 0$,

$$C_n^\lambda(1) = \frac{\Gamma(n+2\lambda)}{n!\Gamma(2\lambda)}; \tag{2.2}$$

for $\lambda = 0$ and $n \geq 1$,

$$C_n^0(1) = \frac{2}{n};$$

for $\lambda = 0$ and $n = 0$,

$$C_n^0(1) = 1;$$

and

$$\left|C_n^\lambda(x)\right| \leq C_n^\lambda(1), \qquad -1 \leq x \leq 1. \tag{2.3}$$

The Gegenbauer polynomials are orthogonal under their weight function $(1-x^2)^{\lambda-\frac{1}{2}}$:

$$\int_{-1}^{1} (1-x^2)^{\lambda-\frac{1}{2}} C_k^\lambda(x) C_n^\lambda(x) dx = \delta_{k,n} h_n^\lambda$$

where, for $\lambda > 0$,

$$h_n^\lambda = \pi^{\frac{1}{2}} C_n^\lambda(1) \frac{\Gamma(\lambda+\frac{1}{2})}{\Gamma(\lambda)(n+\lambda)}; \tag{2.4}$$

for $\lambda = 0$ and $n \geq 1$,

$$h_n^0 = \frac{2\pi}{n^2};$$

for $\lambda = 0$ and $n = 0$,

$$h_0^0 = \pi.$$

We will need to use the Stirling's formula and the estimate of h_n^λ for the asymptotics of the Gegenbauer polynomials for large n and λ.

Lemma 2.2 *We have the Stirling's formula*

$$(2\pi)^{\frac{1}{2}} x^{x+\frac{1}{2}} e^{-x} \le \Gamma(x+1) \le (2\pi)^{\frac{1}{2}} x^{x+\frac{1}{2}} e^{-x+\frac{1}{12x}}, \qquad x \ge 1. \tag{2.5}$$

Lemma 2.3 *There exists a constant C independent of λ and n such that*

$$C^{-1}\frac{\lambda^{\frac{1}{2}}}{n+\lambda} C_n^\lambda(1) \le h_n^\lambda \le C\frac{\lambda^{\frac{1}{2}}}{n+\lambda} C_n^\lambda(1). \tag{2.6}$$

We would need to estimate $||\frac{d^t}{dx^t}\frac{d^l}{dy^l}\{(1-y^2)^{l+\lambda-\frac{1}{2}}\frac{dy}{dx}\}||_{L^\infty}$, therefore we need the following preliminaries first.

Remark 2.4 $\frac{d^l}{dy^l}\{(1-y^2)^{l+\lambda-\frac{1}{2}}\frac{dy}{dx}\}$ *has up to t-th derivatives in x, where $t = \lfloor \frac{\lambda+\frac{1}{2}}{q} \rfloor - 1$, the largest integer below $\frac{\lambda+\frac{1}{2}}{q} - 1$.*

It is easy to observe that

$$\frac{d^n}{dx^n}(1-y(x)^2)^{l+\lambda-\frac{1}{2}} = A^n Y_1^n Y_2^n Y_3^n, \qquad 0 \le n \le t+1 \tag{2.7}$$

where

$$A = \frac{2^q}{2q}, \qquad Y_1^n = (1-y(x)^2)^{l+\lambda-\frac{1}{2}-qn}, \qquad Y_2^n = (1-y(x))^{n(q-1)},$$

and Y_3^n satisfies the following recursive relation:

$$\begin{aligned} Y_3^0 &= 1 \\ Y_3^{n+1} &= -\left[(2l+2\lambda-qn-n-1)y + n(q-1)\right] Y_3^n + (1-y^2)\frac{d}{dy}Y_3^n, \quad 0 \le n \le t. \end{aligned}$$

It is easy to show that Y_3^n is an n-th degree polynomial of y. We have the following estimate on Y_3^n.

Lemma 2.5 *We have, for $0 \le n \le t+1$,*

$$|Y_3^n| \le (2l+2\lambda)^n, \qquad y \in [-1,1] \tag{2.8}$$

Proof : The proof can be found in [2]. In the proof, Y_3^n is rewritten as $Y_3^n = \sum_{i=0}^n a_i y^i$ and we denote $S_n = \sum_{i=0}^n |a_i|$. The proof also provide an estimate for S_n:

$$S_n \le (2l + 2\lambda)^n \tag{2.9}$$

This will be used later in the estimation.

∎

For $l \ge 1$,

$$\frac{d^t}{dx^t}\left\{\frac{d^l}{dy^l}(1-y^2(x))^{l+\lambda-\frac{1}{2}}\frac{dy}{dx}\right\} = \frac{d^{l-1}}{dy^{l-1}}\frac{d^{t+1}}{dx^{t+1}}\left\{(1-y^2(x))^{l+\lambda-\frac{1}{2}}\right\} \tag{2.10}$$

Lemma 2.6 *We have the following estimate for $l \ge 1$,*

$$\left|\frac{d^{l-1}}{dy^{l-1}}\frac{d^{t+1}}{dx^{t+1}}\left\{(1-y^2(x))^{l+\lambda-\frac{1}{2}}\right\}\right| \le CA^{t+1}2^{l+\lambda}(l+\lambda)^{l+t}, \qquad y \in [-1,1] \tag{2.11}$$

Proof :

$$\frac{d^{l-1}}{dy^{l-1}}\frac{d^{t+1}}{dx^{t+1}}\left\{(1-y^2(x))^{l+\lambda-\frac{1}{2}}\right\} \;=\; A^{t+1}\frac{d^{l-1}}{dy^{l-1}}\{Y_1^{t+1}Y_2^{t+1}Y_3^{t+1}\}$$

For simplicity, we denote

$$\frac{d^i}{dy^i}\{Y_1^{t+1}Y_2^{t+1}Y_3^{t+1}\} = X_1^i X_2^i, \qquad 0 \le i \le l-1$$

where,

$$X_1^i = (1-y^2)^{l+\lambda-\frac{1}{2}-q(t+1)-i}$$

and X_2^i satisfies the following recursive relationship:

$$\begin{aligned} X_2^0 &= Y_2^{t+1}Y_3^{t+1} \\ X_2^{i+1} &= -\left[2(l+\lambda-q(t+1)-\frac{1}{2}-i)y\right]X_2^i + (1-y^2)\frac{d}{dy}X_2^i, \quad 0 \le i < l-2. \end{aligned}$$

It is easy to find out that X_2^i is a polynomial of degree $(t+1)q+i$. We only need to prove that

$$|X_2^{l-1}| \leq C2^{l+\lambda}(l+\lambda)^{l+t} \tag{2.12}$$

We can rewrite it as $X_2^i = \sum_{j=0}^{(t+1)q+i} \beta_j^i y^j$. In order to get the upper bound for X_2^{l-1} as in (2.12), we need to estimate X_2^0 first. Let us rewrite $Y_3^{t+1} = \sum_{i=0}^{t+1} a_i y^i$ with its coefficients a_i.

$$\begin{aligned} X_2^0 &= (1-y)^{(t+1)(q-1)} Y_3^{t+1} \\ &= \sum_{j=0}^{(t+1)(q-1)} \binom{(t+1)(q-1)}{j} (-y)^j \sum_{i=0}^{t+1} a_i y^i \\ &= \sum_{j=0}^{(t+1)(q-1)} \sum_{i=0}^{t+1} \binom{(t+1)(q-1)}{j} (-1)^j a_i y^{i+j} \end{aligned}$$

Then we measure the sum of the coefficients of X_2^0,

$$\begin{aligned} \sum_{k=0}^{(t+1)q} |\beta_k^0| &\leq \sum_{j=0}^{(t+1)(q-1)} \sum_{i=0}^{t+1} \binom{(t+1)(q-1)}{j} |a_i| \\ &= \sum_{j=0}^{(t+1)(q-1)} \binom{(t+1)(q-1)}{j} \sum_{i=0}^{t+1} |a_i| \\ &= 2^{(t+1)(q-1)} S_{t+1} \\ &\leq 2^{(t+1)q}(l+\lambda)^{t+1} \end{aligned}$$

Using induction, we can easily get similar result for X_2^{l-1},

$$\sum_{k=0}^{(t+1)q+l-1} |\beta_k^{l-1}| \leq 2^{(t+1)q+l-1}(l+\lambda)^{t+l}$$

which implies that

$$\begin{aligned} |X_2^{l-1}| &\leq 2^{(t+1)q+l-1}(l+\lambda)^{t+l} \\ &\leq C2^{\lambda+l}(l+\lambda)^{t+l} \end{aligned}$$

Thus, we complete the proof. ∎

Remark 2.7 *The result in Lemma 2.6 is also true for $l = 0$. Hence we have*

$$\left\| \frac{d^t}{dx^t} \left\{ \frac{d^l}{dy^l} (1 - y^2(x))^{l+\lambda-\frac{1}{2}} \frac{dy}{dx} \right\} \right\|_{L^\infty} \leq C A^{t+1} 2^{l+\lambda} (l+\lambda)^{l+t}, \qquad l \geq 0 \tag{2.13}$$

3 Truncation Error

In this section, we will establish the error estimate for replacing the Gegenbauer coefficients $\hat{f}^\lambda(l)$ by $\hat{g}^\lambda(l)$ in the Gegenbauer expansion.

Consider the function in the form of

$$f(x) = a(x) + b(x)(1+x)^s$$

where s is a given constant $0 < s = \frac{p}{q} < 1$ with relatively prime positive integers p and q, and $a(x)$ and $b(x)$ are analytic functions satisfying Assumption 1.1.

We assume that the Fourier coefficients $\tilde{f}_n$ $(-N \leq n \leq N)$ are given. Thus we have its truncated Fourier series

$$P_N(f)(x) = \sum_{|n| \leq N} \tilde{f}_n e^{in\pi x}.$$

We are interested in recovering the first m coefficients in the Gegenbauer expansion of $f(x)$. For the function $\bar{f}(y) = f(x(y))$, we have the usual Gegenbauer expansion with the basis $\{C_l^\lambda(y)\}$:

$$f(x(y)) = \bar{f}(y) = \sum_{l=0}^{\infty} \hat{f}^\lambda(l) C_l^\lambda(y(x))$$

where the Gegenbauer coefficients $\hat{f}^\lambda(l)$ are given by (1.12).

Our goal here is to find a good approximation to the first $m \sim N$ Gegenbauer coefficients $\hat{f}^\lambda(l)$ in this expansion. The candidate for approximating the Gegenbauer coefficients $\hat{f}^\lambda(l)$ is:

$$\hat{g}^\lambda(l) = \frac{1}{h_l^\lambda} \int_{-1}^{1} (1-y^2)^{\lambda-\frac{1}{2}} P_N(f) \circ x(y) C_l^\lambda(y) dy. \tag{3.1}$$

Definition 3.1 *The truncation error is defined as*

$$TE(\lambda, m, N) = \max_{-1 \leq y \leq 1} \left| \sum_{l=0}^{m} (\hat{f}^\lambda(l) - \hat{g}^\lambda(l)) C_l^\lambda(y) \right| \tag{3.2}$$

where, $\hat{f}^\lambda(l)$, $\hat{g}^\lambda(l)$ *are defined in (1.12) and (3.1)*

Lemma 3.2 *The truncation error is bounded by*

$$TE(\lambda, m, N) \leq \frac{C}{(N\pi)^{t-1}} \sum_{l=0}^{m} \frac{C_l^\lambda(1)}{h_l^\lambda} \left\| \frac{d^t}{dx^t} M(y) \right\|_{L^\infty}$$

where $t = \lfloor \frac{\lambda+\frac{1}{2}}{q} \rfloor - 1$, *and*

$$M(y) = (1-y^2)^{\lambda-\frac{1}{2}} C_l^\lambda(y) \frac{dy}{dx}$$

Proof We have

$$\begin{aligned}
TE(\lambda, m, N) &= \max_{-1\leq y\leq 1} \left| \sum_{l=0}^{m} \frac{C_l^\lambda(y)}{h_l^\lambda} \int_{-1}^{1} (1-y^2)^{\lambda-\frac{1}{2}} (f - P_N f) \circ x(y) C_l^\lambda(y) dy \right| \\
&\leq \sum_{l=0}^{m} \frac{C_l^\lambda(1)}{h_l^\lambda} \left| \int_{-1}^{1} (1-y^2)^{\lambda-\frac{1}{2}} (f - P_N f) \circ x(y) C_l^\lambda(y) dy \right| \\
&= \sum_{l=0}^{m} \frac{C_l^\lambda(1)}{h_l^\lambda} \left| \int_{-1}^{1} (1-y^2)^{\lambda-\frac{1}{2}} \sum_{|n|>N} \tilde{f}_n e^{in\pi x} C_l^\lambda(y) dy \right| \\
&= \sum_{l=0}^{m} \frac{C_l^\lambda(1)}{h_l^\lambda} \left| \sum_{|n|>N} \tilde{f}_n \int_{-1}^{1} e^{in\pi x} M(y) dx \right| \\
&= \sum_{l=0}^{m} \frac{C_l^\lambda(1)}{h_l^\lambda} \left| \sum_{|n|>N} \frac{\tilde{f}_n}{(in\pi)^t} \int_{-1}^{1} e^{in\pi x} \frac{d^t}{dx^t} M(y) dx \right| \\
&\leq C \sum_{l=0}^{m} \frac{C_l^\lambda(1)}{h_l^\lambda} \sum_{|n|>N} \frac{1}{(n\pi)^t} \left\| \frac{d^t}{dx^t} M(y) \right\|_{L^\infty} \\
&\leq \frac{C}{(N\pi)^{t-1}} \sum_{l=0}^{m} \frac{C_l^\lambda(1)}{h_l^\lambda} \left\| \frac{d^t}{dx^t} M(y) \right\|_{L^\infty}
\end{aligned}$$

The definitions of $\hat{f}^\lambda(l)$ in (1.12) and $\hat{g}^\lambda(l)$ in (3.1) are used in the first equality, (2.3) is used in the second inequality, the definition of the approximation error in (1.6) is used in the third equality, and the substitution (1.11) has been made in the integral in the fourth equality; in the fifth equality, we use integration by parts t times, and the fact that $\frac{d^i}{dx^i} M(y)$ vanishes at $y = \pm 1$ for $0 \leq i \leq t-1$; since $f(x)$ is an L^2-function, its Fourier coefficients $\tilde{f}_n$ are uniformly bounded,i.e., $|\tilde{f}_n| \leq C$, which is used in the sixth inequality. ∎

Lemma 3.3

$$TE(\lambda, m, N) \leq C\frac{(m+1)2^{\lambda}A^{t+1}(m+\lambda)^{m+t+1}\Gamma(\lambda)\Gamma(m+2\lambda)}{m!\Gamma(2\lambda)\Gamma(m+\lambda+\frac{1}{2})(N\pi)^{t-1}}$$

Proof From Lemma 3.2, we have

$$\begin{aligned}
TE(\lambda, m, N) &\leq \frac{C}{(N\pi)^{t-1}}\sum_{l=0}^{m}\frac{C_l^{\lambda}(1)}{h_l^{\lambda}}\left\|\frac{d^t}{dx^t}\left[(1-y^2)^{\lambda-\frac{1}{2}}C_l^{\lambda}(y)\frac{dy}{dx}\right]\right\|_{L^{\infty}} \\
&= \frac{C}{(N\pi)^{t-1}}\sum_{l=0}^{m}\frac{C_l^{\lambda}(1)G(\lambda,l)}{2^l l! h_l^{\lambda}}\left\|\frac{d^t}{dx^t}\left\{\frac{d^l}{dy^l}(1-y^2(x))^{l+\lambda-\frac{1}{2}}\frac{dy}{dx}\right\}\right\|_{L^{\infty}} \\
&\leq C\frac{A^{t+1}2^{\lambda}}{(N\pi)^{t-1}}\sum_{l=0}^{m}\frac{C_l^{\lambda}(1)G(\lambda,l)(l+\lambda)^{l+t}}{l!h_l^{\lambda}} \\
&= C\frac{A^{t+1}2^{\lambda}\Gamma(\lambda)}{\Gamma(2\lambda)(N\pi)^{t-1}}\sum_{l=0}^{m}\frac{\Gamma(l+2\lambda)(l+\lambda)^{l+t+1}}{l!\Gamma(l+\lambda+\frac{1}{2})} \\
&\leq C\frac{A^{t+1}2^{\lambda}\Gamma(\lambda)(m+1)\Gamma(m+2\lambda)(m+\lambda)^{m+t+1}}{m!\Gamma(2\lambda)\Gamma(m+\lambda+\frac{1}{2})(N\pi)^{t-1}}
\end{aligned}$$

where the definition of Gegenbauer polynomial in (2.1) is used in the second step, Lemma 2.6 and Remark 2.7 are used in the third inequality, (2.1) and (2.4) are used in fourth equality; in the last step, we use the fact that $\Gamma(l+2\lambda)(l+\lambda)^{l+t+1}/\{l!\Gamma(l+\lambda+\frac{1}{2})\}$ is an increasing function of l. ∎

Theorem 3.4 *(The exponential decay of the truncation error) Let $\lambda = \alpha N$, $m = \beta N$ with $0 < \alpha, \beta < 1$, then*

$$TE(\alpha N, \beta N, N) \leq CN^2 q_T^N$$

with

$$q_T = \frac{A^{\frac{\alpha}{q}}e^{\beta}(2\alpha+\beta)^{2\alpha+\beta}}{\pi^{\frac{\alpha}{q}}(2\alpha)^{\alpha}\beta^{\beta}(\alpha+\beta)^{(1-\frac{1}{q})\alpha}}.$$

When we choose $\beta = \gamma\alpha$, i.e. $m = \gamma\lambda$, we have

$$q_T = \left(\frac{A^{\frac{1}{q}}e^{\gamma}(2+\gamma)^{2+\gamma}\alpha^{\frac{1}{q}}}{2\pi^{\frac{1}{q}}\gamma^{\gamma}(1+\gamma)^{1-\frac{1}{q}}}\right)^{\alpha}.$$

If we choose α to satisfy

$$\alpha < \frac{\pi}{A}\left(\frac{2\gamma^{\gamma}(1+\gamma)^{1-\frac{1}{q}}}{e^{\gamma}(2+\gamma)^{2+\gamma}}\right)^{q}, \tag{3.3}$$

then $q_T < 1$.

Proof From lemma 3.3, we use Stirling's formula (2.5) and get

$$\begin{aligned}
TE(\lambda, m, N) &\leq C\frac{(m+1)2^{\lambda}A^{t+1}(m+\lambda)^{m+t+1}\Gamma(\lambda)\Gamma(m+2\lambda)}{m!\Gamma(2\lambda)\Gamma(m+\lambda+\frac{1}{2})(N\pi)^{t-1}} \\
&\leq C\frac{e^{m}A^{t+1}(m+2\lambda)^{m+2\lambda-\frac{1}{2}}}{2^{\lambda}\lambda^{\lambda}(m+\lambda)^{\lambda-t-1}m^{m-\frac{1}{2}}(N\pi)^{t-1}}
\end{aligned}$$

hence

$$\begin{aligned}
TE(\alpha N, \beta N, N) &\leq CN^{2}\left(\frac{A(\alpha+\beta)}{\pi}\right)^{t}\left\{\frac{e^{\beta}(2\alpha+\beta)^{2\alpha+\beta}}{(2\alpha)^{\alpha}\beta^{\beta}(\alpha+\beta)^{\alpha}}\right\}^{N} \\
&\leq CN^{2}\left(\frac{A(\alpha+\beta)}{\pi}\right)^{\frac{\alpha}{q}N}\left\{\frac{e^{\beta}(2\alpha+\beta)^{2\alpha+\beta}}{(2\alpha)^{\alpha}\beta^{\beta}(\alpha+\beta)^{\alpha}}\right\}^{N} \\
&= CN^{2}q_{T}^{N}
\end{aligned}$$

where we have used $\frac{\lambda}{q}-2 \leq t = \lfloor\frac{\lambda+\frac{1}{2}}{q}\rfloor - 1 \leq \frac{\lambda}{q}$ in the second step.

∎

4 Regularization Error

Regularization error, the second part of the post-processing error, is caused by approximating $f(x) = a(x) + b(x)(1+x)^{s}$ on $[-1, 1]$ by its Gegenbauer expansion based on the Gegenbauer polynomials $C_{l}^{\lambda}(x(y))$. This has been studied in [2]. We will just quote the result.

The function $\bar{f}(y) = f(x(y)) = a \circ x(y) + b \circ x(y)(1+y)^{p}2^{s-p}$of the variable y is analytic on $y \in [-1, 1]$. Let us consider the Gegenbauer partial sum of the first m terms for the function $f(x)$ given by:

$$f^{\lambda,m}(x) = \sum_{l=0}^{m}\hat{f}^{\lambda}(l)C_{l}^{\lambda}(y(x)), \tag{4.1}$$

with the Gegenbauer coefficients $\hat{f}^{\lambda}(l)$ defined by (1.12).

Definition 4.1 *The regularization error is defined by*

$$RE(\lambda, m) = \max_{-1\leq x\leq 1}\left|f(x) - \sum_{l=0}^{m}\hat{f}^{\lambda}(l)C_{l}^{\lambda}(y(x))\right|$$

$$= \max_{-1\le y\le 1} \left| f(x(y)) - \sum_{l=0}^{m} \hat{f}^{\lambda}(l) C_l^{\lambda}(y) \right| .$$

When both λ and m grow linearly with N, we have the following result for the estimate of regularization error [2].

Theorem 4.2 *(The exponential decay of the regularization error) For the function $f(x) = a(x) + b(x)(1+x)^s$, with analytic functions $a(x)$ and $b(x)$ satisfying Assumption 1.1, if we assume $\lambda = \alpha N$ and $m = \gamma\lambda$, then*

$$\max_{-1\le x\le 1} \left| f(x) - \sum_{l=0}^{m} \hat{f}^{\lambda}(l) C_l^{\lambda}(y(x)) \right| \le C q_R^N$$

where

$$q_R = \left(\frac{(\gamma+2)^{\gamma+2}}{\rho^{\gamma} 2^{\gamma+2} (\gamma+1)^{\gamma+1}} \right)^{\alpha}$$

which is always less than 1.

Proof The proof of this theorem can be found in [2]. ∎

We can combine the estimates for the truncation error and the regularization error and obtain the following theorem about the reconstruction error:

Theorem 4.3 *(Removal of the Gibbs Phenomenon) Consider a function in the form of $f(x) = a(x) + b(x)(1+x)^s$, with given fractional constant $0 < s = \frac{p}{q} < 1$, and $a(x)$ and $b(x)$ are analytic functions satisfying Assumption 1.1. Assume that the Fourier coefficient*

$$\tilde{f}_n = \frac{1}{2} \int_{-1}^{1} f(x) e^{-ik\pi x} dx$$

are known for $-N \le n \le N$. Let $\hat{g}^{\lambda}(l)$, with $0 \le l \le m$, be the Gegenbauer expansion coefficients of $f_N(x) = \sum_{|n|\le N} \hat{f}_n e^{in\pi x}$, defined in (3.1). Then for $\lambda = \alpha N$ and $m = \gamma\lambda$ with

$$\alpha < \frac{\pi}{A} \left(\frac{2\gamma^{\gamma}(1+\gamma)^{1-\frac{1}{q}}}{e^{\gamma}(2+\gamma)^{2+\gamma}} \right)^{q},$$

we have

$$\max_{-1\le x\le 1}\left|f(x)-\sum_{l=0}^{m}\hat{g}^{\lambda}(l)C_l^{\lambda}(y(x))\right|\le C\left(q_T^N+q_R^N\right)$$

where

$$q_T=\left(\frac{A^{\frac{1}{q}}e^{\gamma}(2+\gamma)^{2+\gamma}\alpha^{\frac{1}{q}}}{2\pi^{\frac{1}{q}}\gamma^{\gamma}(1+\gamma)^{1-\frac{1}{q}}}\right)^{\alpha}<1,\qquad q_R=\left(\frac{(\gamma+2)^{\gamma+2}}{\rho^{\gamma}2^{\gamma+2}(\gamma+1)^{\gamma+1}}\right)^{\alpha}<1.$$

Remark 4.4 *In the proof, no attempt has been made to optimize the parameters.*

5 Error analysis for Fourier Galerkin methods with reconstructions

In this section, we will consider the error for Fourier Galerkin methods applied to solve the problem (1.1). The coefficients $a_k(t)$ are determined by requiring that the residual

$$R_N(x,t)=\frac{\partial u_N(x,t)}{\partial t}+c\frac{\partial u_N(x,t)}{\partial x} \tag{5.1}$$

is orthogonal to $\hat{B}_N$. If we rewrite the residual in terms of the Fourier series,

$$R_N(x,t)=\sum_{|k|\le\infty}\hat{R}_k(t)e^{ik\pi x}, \tag{5.2}$$

the orthogonality requirement yields

$$\tilde{R}_k(t)=\frac{1}{2}\int_{-1}^{1}R_N(x,t)e^{-ik\pi x}dx=0,\qquad -N\le k\le N. \tag{5.3}$$

This requirement provides $(2N+1)$ ordinary differential equations to determine the $(2N+1)$ unknowns $a_k(t)$,

$$\frac{d}{dt}a_k(t)+ik\pi c a_k(t)=0,\qquad -N\le k\le N, \tag{5.4}$$

and the corresponding initial conditions are

$$u_N(x,0)=\sum_{|k|\le N}a_k(0)e^{ik\pi x}, \tag{5.5}$$

$$a_k(0)=\frac{1}{2}\int_{-1}^{1}u(x,0)e^{-ik\pi x}dx \tag{5.6}$$

For this linear constant coefficient problem, $R_N(x,t)$ itself is in the space $\hat{B}_N$, therefore the orthogonal complement must be zero, i.e.,

$$R_N(x,t) = 0.$$

It means that we obtain the exact solution of this linear problem by Fourier Galerkin methods. Therefore, the coefficients $a_k(t)$ are exactly the Fourier coefficients for the solution $u(x,t)$, which has singularities at $(2n-1+ct, t)$ (here, $n \in Z$).

In order to recover the exponential accuracy of the solution, we need to take post-processing on its translation $v(x,t) = u(x+ct,t)$, which has a singularity $x = -1$ in the interval $[-1,1]$. And $v(x,t)$ behaviors like $(1+x)^s$ near the singularity. Under this translation, the Fourier coefficients $b_k(t)$ of $v(x,t)$ are

$$b_k(t) = e^{ik\pi ct} a_k(t). \tag{5.7}$$

From the truncated Fourier series $v_N(x,t) = \sum_{|k|\le N} b_k(t) e^{ik\pi x}$, we get a reconstructed approximation

$$v_N^{\lambda,m}(x,t) = \sum_{l=0}^{m} \hat{v}_l^\lambda C_l^\lambda(y(x)), \qquad x \in [-1,1] \tag{5.8}$$

with

$$\hat{v}_l^\lambda = \frac{1}{h_l^\lambda} \int_{-1}^{1} (1-y^2)^{\lambda-\frac{1}{2}} v_N(x(y),t) C_l^\lambda(y) dy. \tag{5.9}$$

Then the numerical solution after processing $u_N^{\lambda,m}(x,t)$ is obtained by

$$u_N^{\lambda,m}(x,t) = Q_N u_N(x,t) = v_N^{\lambda,m}(x-ct,t). \tag{5.10}$$

Combining Theorem 4.3, we obtain the following theorem on the accuracy in spectral methods solving the problem (1.1).

Theorem 5.1 *Using Fourier Galerkin method to solve problem (1.1), the solution $u_N(x,t)$ is defined by*

$$u_N(x,t) = \sum_{|k|\le N} a_k(t) e^{ik\pi x},$$

where the coefficients $a_k(t)$ are determined by (5.4) and (5.6). After postprocessing, we get

$$Q_N u_N(x,t) = v_N^{\lambda,m}(x-ct,t)$$

where, $v_N^{\lambda,m}(x,t)$ is defined in (5.8) and (5.9). The error

$$\max_{-1\leq x\leq 1} |u(x,t) - Q_N u_N(x,t)|$$

is exponentially small, when the parameters λ and m chosen in the post-processing are proportional to N and satisfy (3.3).

6 Numerical results

In this section, we give numerical examples to illustrate the result.

Example 6.1 *We apply the Fourier Galerkin method to solve problem*

$$\begin{cases} u_t + u_x = 0, & x \in [-1,1], \qquad t > 0 \\ u(x,0) = \cos(x) + \sin(x)\sqrt{1+x}, & x \in [-1,1] \end{cases} \tag{6.1}$$

and try to recover the pointwise values of the solution over $[-1,1]$. In the post-processing procedure, the parameters are chosen according to

$$\lambda = \frac{1}{16}N, \qquad m = \frac{3}{80}N. \tag{6.2}$$

For the sake of easy computation, we choose λ and m to be the biggest integer satisfying (6.2).

In Fig. 1(a), we show the errors in the logarithm scale, for N = 40, 80, 160, 320 and 640. In Table 6.1, we show the maximum errors of the reconstructed $v_N^{\lambda,m}(x,t)$ (defined by (5.8) and 5.9) over $x \in [-1,1]$, for each N and the orders of convergence. We can clearly find out that the error is exponentially decaying.

Example 6.2 *We apply the Fourier Galerkin method to solve problem*

$$\begin{cases} u_t + u_x = 0, & x \in [-1,1], \qquad t > 0 \\ u(x,0) = \cos(x) + \sin(x)(1+x)^{\frac{1}{3}}, & x \in [-1,1] \end{cases} \tag{6.3}$$

Table 6.1: Maximum error table

N	L^∞ error	order	λ	m
40	5.91E-001		2	1
80	2.95E-001	1.00	5	3
160	1.56E-002	4.25	10	6
320	1.33E-005	10.19	20	12
640	6.46E-012	20.98	40	24

and try to recover the pointwise values of the solution over $[-1, 1]$*. In the post-processing procedure, the parameters are chosen according to*

$$\lambda = \frac{1}{8}N, \qquad m = \frac{1}{32}N. \tag{6.4}$$

For the sake of easy computation, we choose λ *and* m *to be the biggest integer satisfying (6.4).*

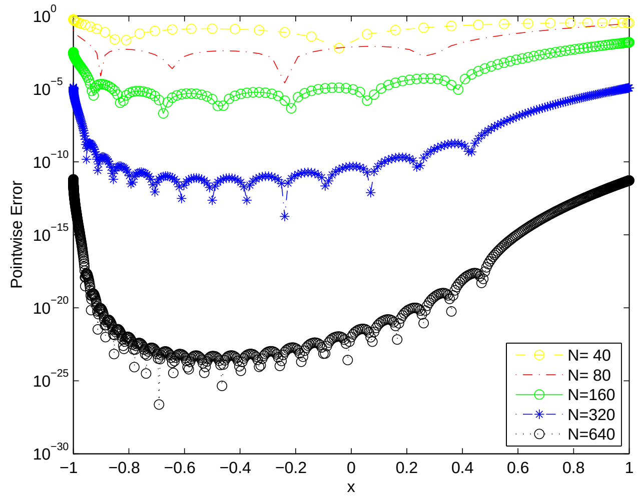
In Fig. 1(b), we show the errors in the logarithm scale, for $N = 40, 80, 160, 320$ *and* 640*. In Table 6.2, we show the maximum errors of the reconstructed* $v_N^{\lambda,m}(x,t)$ *(defined by (5.8) and 5.9) over* $x \in [-1, 1]$*, for each* N *and the orders of convergence. We again clearly find out that the error is exponentially decaying.*

Table 6.2: Maximum error table

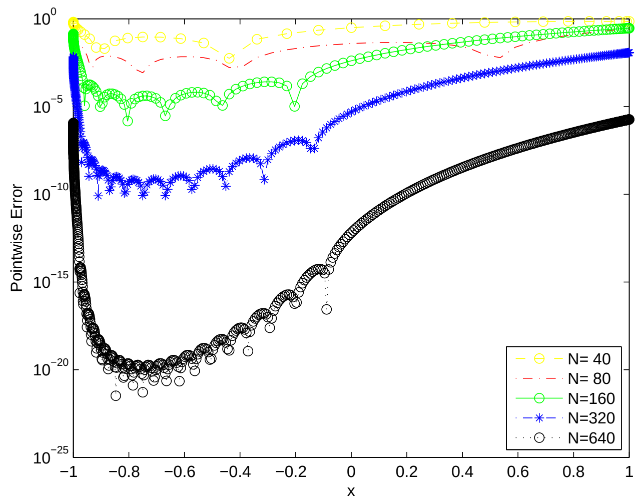
N	L^∞ error	order	λ	m
40	7.13E-001		5	1
80	3.64E-001	0.97	10	2
160	2.91E-001	0.33	20	5
320	1.17E-002	4.64	40	10
640	1.83E-006	12.64	80	20

7 Concluding remarks

We have built the Gegenbauer polynomial based technique to reconstruct approximations with exponential accuracy in the maximum norm, based on the Fourier coefficients, for



(a) Example 1 ($\lambda = \frac{1}{16}N$, $m = \frac{3}{80}N$)



(b) Example 2 ($\lambda = \frac{1}{8}N$, $m = \frac{1}{32}N$)

Figure 6.1: Pointwise errors in the logarithm scale.

piecewise analytic functions where in each piece the function is analytic only in the open interval with unbounded derivative end point singularities. This technique provides a post-processing method for spectral methods to solve transport equations involving such functions to achieve exponential accuracy. Numerical results are provided to demonstrate the theory. This technique has potential applications for solving other partial differential equations whose solutions have such singularities.

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